

Laurinne J Balstad
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Positions

2026- **University of British Columbia**
Postdoctoral researcher, King lab

Education

2025 **University of California, Davis**
PhD in Population Biology
Advisor: Marissa Baskett
Thesis: Facilitating, Promoting, and Impeding Evolutionary Rescue in the Anthropocene

2021 **St. Olaf College, Bachelor of Arts**
Mathematics and Biology

Publications

- 7 **Balstad, Laurinne***, Brennan, Joeseeph*, [12 authors]. **2026**. Eight quick tips for data-model integration. *Accepted at PLoS Computational Biology*. *Shared first authorship.
- 6 Urban, Mark and **Balstad, Laurinne**. **2026**. *Perspective: Evolution to the rescue. Science* 391, 6790. [Invited]
- 5 **Balstad, Laurinne**, Baskett, Marissa, Carlson, Stephanie, Cunningham, Curry, & Westley, Peter. **2026**. Revisiting evolutionary rescue in the wild. *Trends in Ecology & Evolution*. 41(8), 762–774.
- 4 Szuwalski, Cody, Alglave, Baptiste, **Balstad, Laurinne**, Olmos, Maxime, Punt, Andre, Richar, Jon, and Veron, Matthieu. **2026**. Density dependence modulates climate change impacts on easter Bering Sea snow crab. *J. of Applied Ecology*, 63(3), e70316.
- 3 **Balstad, Laurinne**, Godwin, Sean, Krkosek, Martin, Lewis, Mark, and Baskett, Marissa. **2025**. Threshold-based disease treatment approach modulates economic, conservation and evolutionary trade-offs in sea louse-salmon aquaculture system. *Theoretical Ecology*, 18(1), 23.
- 2 **Balstad, Laurinne**, Fedewa, Erin, and Szuwalski, Cody. **2024**. Drivers of Bitter Crab Disease occurrence in Eastern Bering Sea snow crab (*Chionoecetes opilio*). *ICES J. of Marine Science*, 81(6), 1073–1083.
- 1 **Balstad, Laurinne**, Binning, Sandra, Craft, Meggan, Zuk, Marlene, and Shaw, Allison. **2021**. Parasite intensity and the evolution of migratory behavior. *Ecology*, e03229.

White papers

- 10 Kinneen, Michael, Goodman, Maurice. C., Sulc, Anna, **Balstad, Laurinne**, ... [19 authors]. **2025**. Status of widow rockfish stock off the U.S. West Coast in 2025. *Pacific Fishery Management Council [peer reviewed]*.
- 9 **Balstad, Laurinne**, Sullivan, Jane, and Monnahan, Cole. **2024**. A Plain Language Guide to Model Validation in REMA. White paper, *North Pacific Fisheries Management Council [peer reviewed]*.
- 8 Pinsky, Malin, ... **Balstad, Laurinne**, ... [17 authors]. **2024**. Evolutionary Resilience for Kelp: Prospects and Challenges. *Workshop Report, Monterey Bay Aquarium, University of California, Santa Cruz*. doi: 10.6084/m9.figshare.26882734

Manuscripts in the pipeline, available upon request

Balstad, Laurinne, and Baskett, Marissa. Drivers of the likelihood of evolutionary rescue in response to novel diseases. *In revision at PNAS*.

Balstad, Laurinne, Ashby, Ben, Godwin, Sean, Krkosek, Martin, Lewis, Mark, and Baskett, Marissa. Predation dynamics shape sublethal virulence evolution in macroparasites. *In prep*.

Jarvis-Cross, Madeline*, **Balstad, Laurinne***, Amy A. Briggs, Matthew Mensinger, Anjana Parandhaman, Eric Palkovacs, Marissa L. Baskett. Mitigating unintended consequences of population supplementation in imperilled host-pathogen systems. **Shared first authorship. Submitted to Conservation Science and Practice*.

Grants & Funding (\$235K)

- 2025 Jastro-Fields Small Grant, *University of California, Davis*, **\$1,500** Stipend
- 2024 Jastro-Fields Small Grant, *University of California, Davis*, **\$3,000** Undergraduate researcher stipend support
- 2023 Jastro-Fields Small Grant, *University of California, Davis*, **\$2,775** Fieldwork travel/fees
- 2021 Graduate Research Fellowship, *National Science Foundation*, **\$144,000**
- 2021 Graduate Research Traineeship, *National Science Foundation, Sustainable Oceans NRT* **\$46,000**
- 2021 Dean's Distinguished Graduate Fellowship, *University of California, Davis*, **\$39,000**

Teaching

- 2025 **Instructor of Record** (co-taught), Population Ecology, *University of California, Davis*
"Seminar on College Teaching" certificate [10 weeks]
- 2024 **Teaching assistant**, Population Ecology, *University of California, Davis*
Discussion leader and course organizer (TA), Marine disease (graduate course), *University of California, Davis*
- 2023 **Teaching assistant**, Marine Ecosystem Research and Management, University of New Hampshire
Guest lectures: Data Management; Ecosystem indicators: Bitter crab disease case study
Discussion leader and course organizer (TA), Ecosystem-based fisheries management (graduate course), *University of California, Davis*

Student research mentorship

- 2026 Post-baccalaureate researcher, "Predicting shifts in emergence hotspots of avian influenza in arctic marine mammals under climate change"
seed2STEM High school student researcher, "Marine disease in fisheries management"
Program to provide research experience to Indigenous high school students
- 2025 Undergraduate researcher, "Reintroductions and evolutionary rescue in host-pathogen systems"
- 2023 Undergraduate researcher, "Patterns of cohort resonance across Pacific coast fish stocks"

Presentations

- Upcoming *Invited seminar*, Center for Fisheries and Ocean Sciences, University of Alaska, Fairbanks

Invited seminar, Maud Menten Institute Seminar Series, hosted at University of Manitoba
Invited seminar, Mathematical Biology Seminar Series, University of Victoria
- 2026 *Talk*, BLISS seminar series, University of British Columbia
Talk, University of British Columbia, Mathematical Biology Symposium Fall 2026

Poster , Ecological Society of America Annual Meeting (Salt Lake City, US)
 co-Session Organizer ("*Vector-borne disease*") at Ecological Society of America Annual Meeting (Salt Lake City, US)
 Invited attendee , NAS-Royal Society Forum on Shifts and tipping points in ocean systems (London, UK)
 Talk , University of British Columbia, Mathematical Biology Symposium Spring 2026
 Talk , Society of Mathematical Biology, Population Dynamics Conference (Online)
 2025 Invited panelist , OneHealth Symposium, University of California, Davis
 Seminar, Center for Population Biology Seminar Series, University of California, Davis
 Talk & co-Session Organizer ("*Evolutionary Rescue*") at Ecological Society of America Annual Meeting (Baltimore, US)
 Talk, Max Planck Evolutionary Rescue Workshop (Plön, Germany)
 2024 Talk, Western Society of Naturalists (Portland, US)
 Seminar, Center for Population Biology Seminar Series, University of California, Davis
 2023 Poster, Western Society of Naturalists (Monterey, US)
 Talk, Alaska Marine Science Symposium (Anchorage, US)
 2021 Talk, SIAM Dynamical Systems (Online)
 2020 Poster, Society of Mathematical Biology Annual Meeting (Online)

Service

2026 **Vice Chair, Disease Ecology Section**, Ecological Society of America
 2025 **Workshop co-leader**, Introduction to R and poster making (9 lesson series), *University of California, Davis Ecology, Evolution, and Working Landscapes REU*
 2025 **Graduate Student Representative Center for Population Biology Postdoctoral Fellowship Committee**, University of California, Davis
 2022-2025 **Co-chair and program leader, Ecology & Evolution Preview Program**, University of California, Davis, *including Co-chair, Mentorship co-coordinator, Workshop presenter*
Program encourages students (n > 150) from underrepresented backgrounds to consider and apply to graduate school in Ecology & Evolution by making application process transparent
 2021-2025 **Student Diversity Committee**, University of California, Davis
 2024 **Professors for the Future**, *University of California, Davis*
Professional development program, with service project "Chat with and editor: Understanding the hidden curriculum of publishing" (workshop series)
 Reviewer *Evolutionary Applications, American Naturalist, Ecosphere, Ecological Applications*

Outreach

2024 **Juneau Seaweed: Modeling Science Bioenergetics for Grades 6-8**
Collaboration with Alaska Fisheries Science Center, NOAA
Shinyapp with lesson plan linked: https://lauriebalstad.shinyapps.io/SAL_GROW/
University of Minnesota: Simulating selection on migration for Grade 12
Collaboration with Martha Torstenson and Allison Shaw
Lesson plan with Shinyapp: <https://qubeshub.org/publications/5103/2>